

ACTIVITY 12 — FACILITATOR GUIDE**Value Stream Map and Takt Time**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Elective Lab	45 min	MEASURE	A3, A4

Purpose of this activity

Quantify flow, lead time and takt time across the value stream (A3, A4).

Extend process mapping into Lean metrics. Map the value stream with process and wait times, calculate process cycle efficiency, then compute takt time to see whether the process can meet customer demand.

What learners produce

A value stream map with process and wait times, a calculated process cycle efficiency and a takt time expressed per unit.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework
- A browser — this activity uses an online tool (URL in the run-sheet).

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55

minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

data/ — the mock data you analyse:

- process-steps.csv — 7 rows (step_no, activity, actor, system, process_time_min, wait_time_min, handoff_to)

templates/ — the worksheets you complete:

- takt-time.csv (input, value)
- value-stream-map.csv (step_no, activity, process_time_min, wait_time_min, VA_or_NVA)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Map the value stream: each step with its process time and the wait time between steps. Start from your Lab 3 process map. Tool: https://alfredang.github.io/sipoc/	Learners forget wait time is where the lead time actually goes — process-steps.csv shows it.
2	Total the value-added time and the lead time, then compute process cycle efficiency.	Process cycle efficiency in a service process is often under 10%; that shock is the lesson.
3	Calculate takt time = available working time / customer demand.	
4	Compare cycle time against takt time to identify the bottleneck step.	

Suggested timing

Minutes	Phase
15 min	Map the value stream with process and wait times
10 min	Total VA time and lead time
10 min	Compute process cycle efficiency
10 min	Calculate takt time and find the bottleneck

Total: 45 minutes

Discussion & decision prompts

1. What proportion of total lead time is actually value-added?
2. Which step is the bottleneck against takt time?
3. Where is the biggest single block of waiting, and who owns it?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the MEASURE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Estimate the process cycle efficiency of one process you run.
- Would your process meet demand if measured against takt time?

Facilitator notes — what to watch for

- Learners forget wait time is where the lead time actually goes — process-steps.csv shows it.
- Process cycle efficiency in a service process is often under 10%; that shock is the lesson.

Success criterion

Your lead time equals the sum of all process and wait times, and takt time is expressed per unit.

Assessment link

This activity builds the MEASURE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.